

Manual

Visualization of Energy Demand HLS-EBV 8.2

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1 Planning Strategy: Energy Demand

Purpose

Application package extending the graphical planning in the HYDRA Shop Floor Scheduling module with planning strategies for energy requirements. These strategies enable you to plan production orders in view of an optimal energy demand. The chart of the HYDRA Shop Floor Scheduling module totals the consumptions of the planned operations for the materials of the ECF groups and a specific point in time. You can also show an upper limit.

Use the function package if:

- you want to view information on energy consumption in the HYDRA Shop Floor Scheduling module.

Integration

In order to use this function package, you also have to use the Energy Management and the HYDRA Shop Floor Scheduling module.

Features

Functions to edit ECF groups, to assign material types to defined ECF groups and to visualize material types (e.g. energy sources) in the HYDRA Shop Floor Scheduling module.

- Use this function to create or modify ECF groups in the system.
- Use this function to assign material types to your configured ECF groups.
- Use this function to visualize material types (e.g. energy sources) in the HYDRA Shop Floor Scheduling module.

2 ECF groups

Overview

HYDRA menu	Master data → Production control → ECF groups
FEDRA menu	Advanced Resource Planning → Master data → ECF groups
Transaction code	ecfgrp
Function authorization	ecfgrp

This document describes the application "ECF groups" of the client.

Purpose

You use this function to create and modify ECF groups in the system.

Integration

ECF groups are used to show energy requirements of operations in the Shop Floor Scheduling.

Selection criteria

The application provides the following selection criteria:

ECF group

Uses the group name to select the ECF group.

Upper limit value

Uses the upper limit value to select the ECF group.

Modified by

Uses the person who last modified the data record to select the ECF group.

Consumption during setup time

Uses the setting *Consumption during setup time* to select the ECF group. In tri-state mode, all groups are selected.

Consumption during processing time

Uses the setting *Consumption during processing time* to select the ECF group. In tri-state mode, all groups are selected.

Field descriptions

ECF group

Name of the ECF group

Consumption during setup time

This field specifies if material types assigned to this ECF group are consumed during setup.

Consumption during processing time

This field specifies if material types assigned to this ECF group are consumed during processing.



Note: When you create or change the ECF groups, at least one of the options "Consumption during setup time" or "Consumption during processing time" must be active. Otherwise you cannot create or change a data record.

Description

Use this field to enter a detailed description of the ECF group.

Upper limit value

This field specifies the upper limit value that is displayed for this ECF group in the graphic planning.

The graphic planning integrates seconds to display the planned energy consumption. We recommend to take this into account when you define the upper limit value.

Example (we used large numbers for demonstration purposes)



- An OP has a target quantity of 1,000 pieces, a processing time of 1:00 hour and an input quantity of 100 kWh that is stored as a component. In this example, the resulting required quantity of the OP is 100,000 kWh (target quantity x input quantity). In the graphic planning, an energy consumption of 27,78 kW is displayed for the OP (required quantity / processing time [sec]).
- In this example, the upper limit value is understood as load limit. If the supplier provides 27,000 kWh within a time frame of 15 minutes, an average of 30 kW is available per second in this time frame (provided quantity / time frame [sec]). Here, you should use 30 kW as upper limit value.

Unit

Unit of the upper limit value

Lower limit value

This field specifies the lower limit value that is displayed for this ECF group in the graphic planning. Currently, this value is set to 0 and cannot be changed.

Unit

Unit of the lower limit value. The unit cannot be changed.

Modified by

Person who last modified the ECF group

Modified on

Point in time of the last change of this ECF group

Toolbar - tab *Main page***Insert**

Opens the dialog to add an ECF group

**Copy**

Opens the dialog to copy an ECF group

**Edit**

Opens the dialog to edit an ECF group

**Delete**

Deletes an ECF group

3 ECF Group Assignment

Overview

HYDRA menu	Master data → Production control → ECF group assignment
FEDRA menu	Advanced Resource Planning → Master data → ECF group assignment
Transaction code	ecfgrpas
Function authorization	ecfgrpas

This document describes the application "ECF group assignment" on the client.

Purpose

You use the function to assign material types to your configured ECF groups.

Integration

In order to visualize the expected energy consumption in the graphic planning, you must assign the relevant material types of the components entered for the operations to ECF groups.

Prerequisite

You have configured the relevant material types and ECF groups in the master data.

Selection criteria

The following selection criteria are available in the application:

ECF group

Selection according to the selected ECF group

Material type

Selection according to the selected material type

Field descriptions

ECF group

Name of the ECF group



Several material types can be assigned to one ECF group.

Description

More detailed description of the ECF group assigned to this group in the application "ECF groups".

Material type

Material type to be assigned to the ECF group.



Each material type may be assigned to a maximum of one ECF group.



For creating and editing a data record, only material types configured in the application "Material types" are available.

Editor

Last editor of the ECF group assignment.

Last modification

Time of last modification of the ECF group assignment.

Toolbar**General tab****Insert**

Opens the dialog to insert an ECF group assignment.

**Copy**

Opens the dialog to copy an ECF group assignment.

**Edit**

Opens the dialog to edit an ECF group assignment.

**Delete**

Deletes an ECF group assignment.